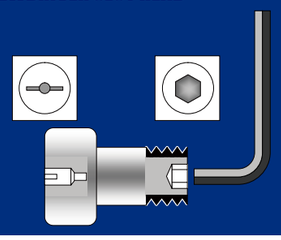
CAM FOLLOWERS

HEXAGON SOCKET ON THREAD SIDE
SCREWDRIVER SLOT HEAD

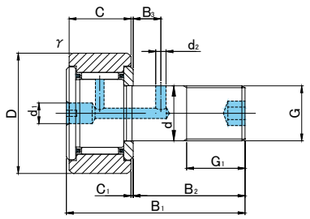


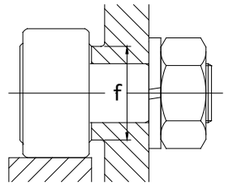


CF..B



CF..VB





CF..B TYPE

Prepacked Grease

Stud diameter (mm)		Designation: Cylindrical outer ring		Designation: Crowned outer ring R500(CF12 ～ CF18) R1000(CF20 ≦)		Dimensions (mm)													Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	B3	C1	d1	d2	H	rs min	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
12	0 -0.018	**CF12B**	**CF12UUB**	**CF12RB**	**CF12UURB**	**30**	14	12	M12×1.5	13	40	25	6	0.6	6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CF12VB	**CF12VUUB**	**CF12VRB**	**CF12VUURB**															13 430	19 700	11 270	6 000				
12	0 -0.018	**CF12-1B**	**CF12-1UUB**	**CF12-1RB**	**CF12-1UURB**	**32**	14	12	M12×1.5	13	40	25	6	0.6	6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CF12-1VB	**CF12-1VUUB**	**CF12-1VRB**	**CF12-1VUURB**															13 430	19 700	11 270	6 000				
16	0 -0.018	**CF16B**	**CF16UUB**	**CF16RB**	**CF16UURB**	**35**	18	16	M16×1.5	17	52	32.5	8	0.8	6	3	6	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CF16VB	**CF16VUUB**	**CF16VRB**	**CF16VUURB**															20 680	37 630	19 800	4 500				
18	0 -0.018	**CF18B**	**CF18UUB**	**CF18RB**	**CF18UURB**	**40**	20	18	M18×1.5	19	58	36.5	8	0.8	6	3	6	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CF18VB	**CF18VUUB**	**CF18VRB**	**CF18VUURB**															25 280	51 350	26 560	3 500				
20	0 -0.021	**CF20B**	**CF20UUB**	**CF20RB**	**CF20UURB**	**52**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	8	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CF20VB	**CF20VUUB**	**CF20VRB**	**CF20VUURB**															33 120	64 480	32 140	3 500				
20	0 -0.021	**CF20-1B**	**CF20-1UUB**	**CF20-1RB**	**CF20-1UURB**	**47**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	8	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CF20-1VB	**CF20-1VUUB**	**CF20-1VRB**	**CF20-1VUURB**															33 120	64 480	32 140	3 500				
24	0 -0.021	**CF24B**	**CF24UUB**	**CF24RB**	**CF24UURB**	**62**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	8	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CF24VB	**CF24VUUB**	**CF24VRB**	**CF24VUURB**															46 550	92 020	49 500	3 000				
24	0 -0.021	**CF24-1B**	**CF24-1UUB**	**CF24-1RB**	**CF24-1UURB**	**72**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	8	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CF24-1VB	**CF24-1VUUB**	**CF24-1VRB**	**CF24-1VUURB**															46 550	92 020	49 500	3 000				
30	0 -0.021	**CF30B**	**CF30UUB**	**CF30RB**	**CF30UURB**	**80**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CF30VB	**CF30VUUB**	**CF30VRB**	**CF30VUURB**															67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-1B**	**CF30-1UUB**	**CF30-1RB**	**CF30-1UURB**	**85**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CF30-1VB	**CF30-1VUUB**	**CF30-1VRB**	**CF30-1VUURB**															67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-2B**	**CF30-2UUB**	**CF30-2RB**	**CF30-2UURB**	**90**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CF30-2VB	**CF30-2VUUB**	**CF30-2VRB**	**CF30-2VUURB**															67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

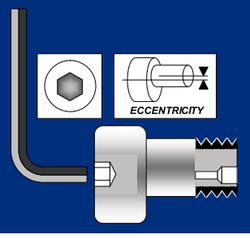
TYPE	Cylindrical outer ring	Crowned outer ring
CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF3/CF4/CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	—	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
HEXAGON SOCKET ON STUD HEAD

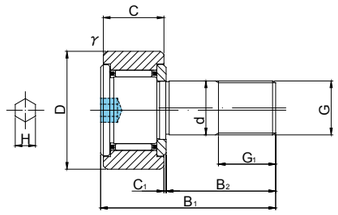




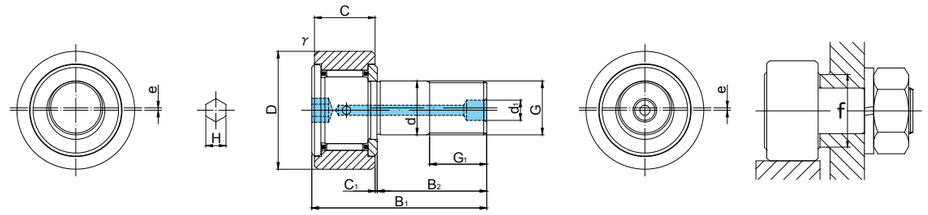
CFH..A



CFH..VA



CFH5(V)A～CFH10−1(V)A



CFH12(V)A～CFH12−1(V)A

CFH..A TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R250(CF5) R500(CF6 ～ CF12-1), Without seals	Crowned outer ring, With seals	D	C	d	G	G1	B1	B2	C1	d1	H	r s min	e	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
5	0 -0.012	CFH 5A	CFH 5UUA	CFH 5RA	CFH 5URA	13	9	5	M5×0.8	7.5	23	13	0.5	—	3	0.3	0.2	9.7	3 140	2 770	1 420	29 000	2 250	780	2	10.5
		CFH 5VA	CFH 5VUUA	CFH 5VRA	CFH 5VURA														5 100	5 500	1 420	11 600				
6	0 -0.012	CFH 6A	CFH 6UUA	CFH 6RA	CFH 6URA	16	11	6	M6×1	8	28	16	0.6	—	3	0.3	0.25	11	3 630	3 630	2 110	25 000	3 430	1 080	3	18.5
		CFH 6VA	CFH 6VUUA	CFH 6VRA	CFH 6VURA														6 960	8 530	2 110	12 000				
8	0 -0.015	CFH 8A	CFH 8UUA	CFH 8RA	CFH 8URA	19	11	8	M8×1.25	10	32	20	0.6	—	4	0.3	0.25	13	4 310	4 710	4 710	20 000	4 020	1 370	8	28.5
		CFH 8VA	CFH 8VUUA	CFH 8VRA	CFH 8VURA														8 130	11 170	4 710	9 000				
10	0 -0.015	CFH10A	CFH10UUA	CFH10RA	CFH10URA	22	12	10	M10×1.25	12	36	23	0.6	—	5	0.3	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CFH10VA	CFH10VUUA	CFH10VRA	CFH10VURA														9 510	14 500	7 450	7 500				
10	0 -0.015	CFH10-1A	CFH10-1UUA	CFH10-1RA	CFH10-1URA	26	12	10	M10×1.25	12	36	23	0.6	—	5	0.3	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CFH10-1VA	CFH10-1VUUA	CFH10-1VRA	CFH10-1VURA														9 510	14 500	7 450	7 500				
12	0 -0.018	CFH12A	CFH12UUA	CFH12RA	CFH12URA	30	14	12	M12×1.5	13	40	25	0.6	6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFH12VA	CFH12VUUA	CFH12VRA	CFH12VURA														13 430	19 700	11 270	6 000				
12	0 -0.018	CFH12-1A	CFH12-1UUA	CFH12-1RA	CFH12-1URA	32	14	12	M12×1.5	13	40	25	0.6	6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFH12-1VA	CFH12-1VUUA	CFH12-1VRA	CFH12-1VURA														13 430	19 700	11 270	6 000				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

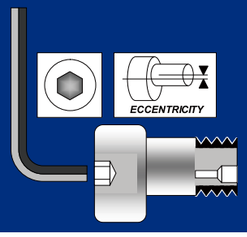
TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	—	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

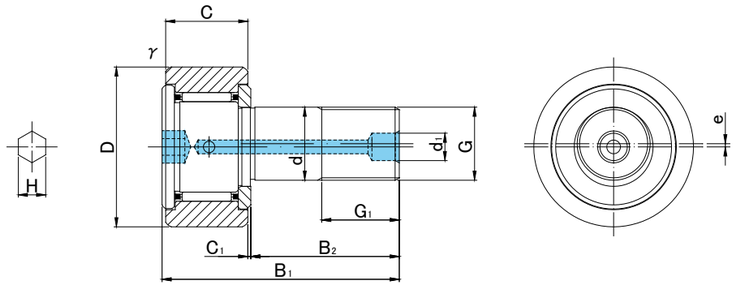
CAM FOLLOWERS

SOLID ECCENTRIC TYPE
HEXAGON SOCKET ON STUD HEAD





CFH..A　　CFH..VA



CFH..A TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R250(CF5) R500(CF6 ~ CF18) R1000(CF20 ≦), Without seals	Crowned outer ring, With seals	D	C	d	G	G1	B1	B2	C1	d1	H	r s min	e	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
16	0 -0.018	CFH16A	CFH16UUA	CFH16RA	CFH16UURA	35	18	16	M16×1.5	17	52	32.5	0.8	6	6	0.6	0.5	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFH16VA	CFH16VUUA	CFH16VRA	CFH16VUURA														20 680	37 630	19 800	4 500				
18	0 -0.018	CFH18A	CFH18UUA	CFH18RA	CFH18UURA	40	20	18	M18×1.5	19	58	36.5	0.8	6	6	1	0.6	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFH18VA	CFH18VUUA	CFH18VRA	CFH18VUURA														25 280	51 350	26 560	3 500				
20	0 -0.021	CFH20A	CFH20UUA	CFH20RA	CFH20UURA	52	24	20	M20×1.5	21	66	40.5	0.8	8	8	1	0.7	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFH20VA	CFH20VUUA	CFH20VRA	CFH20VUURA														33 120	64 480	32 140	3 500				
20	0 -0.021	CFH20-1A	CFH20-1UUA	CFH20-1RA	CFH20-1UURA	47	24	20	M20×1.5	21	66	40.5	0.8	8	8	1	0.7	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFH20-1VA	CFH20-1VUUA	CFH20-1VRA	CFH20-1VUURA														33 120	64 480	32 140	3 500				
24	0 -0.021	CFH24A	CFH24UUA	CFH24RA	CFH24UURA	62	29	24	M24×1.5	25	80	49.5	0.8	8	8	1	0.8	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFH24VA	CFH24VUUA	CFH24VRA	CFH24VUURA														46 550	92 020	49 500	3 000				
24	0 -0.021	CFH24-1A	CFH24-1UUA	CFH24-1RA	CFH24-1UURA	72	29	24	M24×1.5	25	80	49.5	0.8	8	8	1	0.8	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFH24-1VA	CFH24-1VUUA	CFH24-1VRA	CFH24-1VUURA														46 550	92 020	49 500	3 000				
30	0 -0.021	CFH30A	CFH30UUA	CFH30RA	CFH30UURA	80	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFH30VA	CFH30VUUA	CFH30VRA	CFH30VUURA														67 620	144 060	73 700	2 200				
30	0 -0.021	CFH30-1A	CFH30-1UUA	CFH30-1RA	CFH30-1UURA	85	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CFH30-1VA	CFH30-1VUUA	CFH30-1VRA	CFH30-1VUURA														67 620	144 060	73 700	2 200				
30	0 -0.021	CFH30-2A	CFH30-2UUA	CFH30-2RA	CFH30-2UURA	90	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CFH30-2VA	CFH30-2VUUA	CFH30-2VRA	CFH30-2VUURA														67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

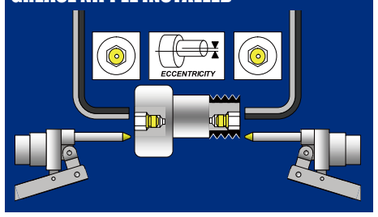
TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	STOP PLUG	RESIN PLUG	NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	—	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
HEXAGON SOCKET ON BOTH SIDES
GREASE NIPPLE INSTALLED

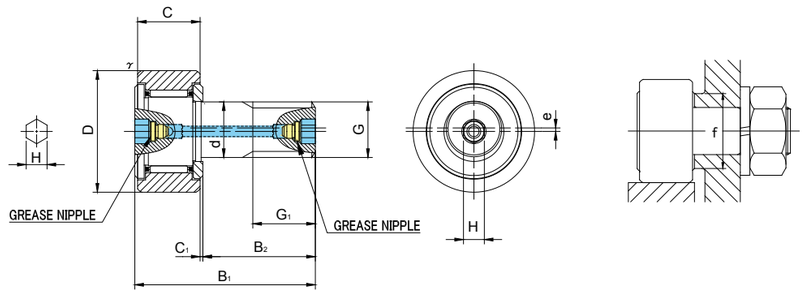




CFH..AB



CFH..VAB



CFH..AB TYPE

Prepacked Grease

Stud diameter (mm)		Designation				Dimensions (mm)											Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
		Cylindrical outer ring		Crowned outer ring R500(CF10 ～ CF18) R1000(CF20 ≦)																				
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	C1	H	r's min	e	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
12	0 -0.018	**CFH12AB**	**CFH12UUAB**	**CFH12RAB**	**CFH12UURAB**	**30**	14	12	M12×1.5	13	40	25	0.6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFH12VAB	**CFH12VUUAB**	**CFH12VRAB**	**CFH12VUURAB**													13 430	19 700	11 270	6 000				
12	0 -0.018	**CFH12-1AB**	**CFH12-1UUAB**	**CFH12-1RAB**	**CFH12-1UURAB**	**32**	14	12	M12×1.5	13	40	25	0.6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFH12-1VAB	**CFH12-1VUUAB**	**CFH12-1VRAB**	**CFH12-1VUURAB**													13 430	19 700	11 270	6 000				
16	0 -0.018	**CFH16AB**	**CFH16UUAB**	**CFH16RAB**	**CFH16UURAB**	**35**	18	16	M16×1.5	17	52	32.5	0.8	6	0.6	0.5	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFH16VAB	**CFH16VUUAB**	**CFH16VRAB**	**CFH16VUURAB**													20 680	37 630	19 800	4 500				
18	0 -0.018	**CFH18AB**	**CFH18UUAB**	**CFH18RAB**	**CFH18UURAB**	**40**	20	18	M18×1.5	19	58	36.5	0.8	6	1	0.6	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFH18VAB	**CFH18VUUAB**	**CFH18VRAB**	**CFH18VUURAB**													25 280	51 350	26 560	3 500				
20	0 -0.021	**CFH20AB**	**CFH20UUAB**	**CFH20RAB**	**CFH20UURAB**	**52**	24	20	M20×1.5	21	66	40.5	0.8	8	1	0.7	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFH20VAB	**CFH20VUUAB**	**CFH20VRAB**	**CFH20VUURAB**													33 120	64 480	32 140	3 500				
20	0 -0.021	**CFH20-1AB**	**CFH20-1UUAB**	**CFH20-1RAB**	**CFH20-1UURAB**	**47**	24	20	M20×1.5	21	66	40.5	0.8	8	1	0.7	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFH20-1VAB	**CFH20-1VUUAB**	**CFH20-1VRAB**	**CFH20-1VUURAB**													33 120	64 480	32 140	3 500				

CFH..AB TYPE has oil holes (grease nipple) which are prepared in the stud head and thread side, also an oil hole is prepared outer surface of stud.

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

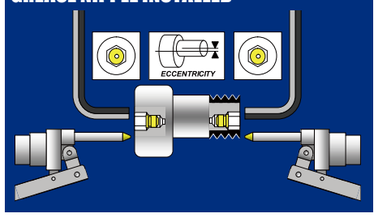
TYPE	Cylindrical outer ring	Crowned outer ring
CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	Grease Nipple	NUT
All types	Installed	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
HEXAGON SOCKET ON BOTH SIDES
GREASE NIPPLE INSTALLED

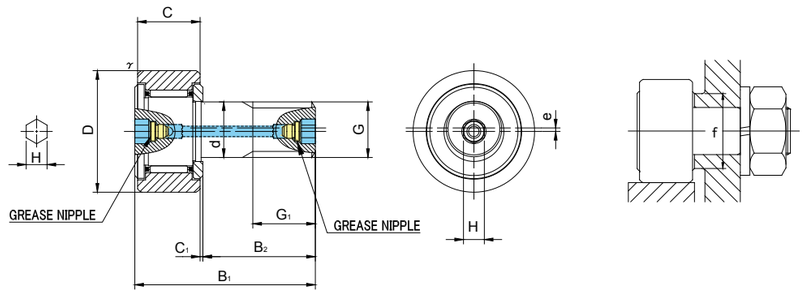




CFH..AB



CFH..VAB



CFH..AB TYPE

Prepacked Grease

Stud diameter (mm)		Designation: Cylindrical outer ring		Designation: Crowned outer ring R1000(CF24 ≦)		Dimensions (mm)											Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	C1	H	r's min	e	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
24	0 -0.021	**CFH24AB**	**CFH24UUAB**	**CFH24RAB**	**CFH24UURAB**	**62**	29	24	M24×1.5	25	80	49.5	0.8	8	1	0.8	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFH24VAB	**CFH24VUUAB**	**CFH24VRAB**	**CFH24VUURAB**													46 550	92 020	49 500	3 000				
24	0 -0.021	**CFH24-1AB**	**CFH24-1UUAB**	**CFH24-1RAB**	**CFH24-1UURAB**	**72**	29	24	M24×1.5	25	80	49.5	0.8	8	1	0.8	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFH24-1VAB	**CFH24-1VUUAB**	**CFH24-1VRAB**	**CFH24-1VUURAB**													46 550	92 020	49 500	3 000				
30	0 -0.021	**CFH30AB**	**CFH30UUAB**	**CFH30RAB**	**CFH30UURAB**	**80**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFH30VAB	**CFH30VUUAB**	**CFH30VRAB**	**CFH30VUURAB**													67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-1AB**	**CFH30-1UUAB**	**CFH30-1RAB**	**CFH30-1UURAB**	**85**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CFH30-1VAB	**CFH30-1VUUAB**	**CFH30-1VRAB**	**CFH30-1VUURAB**													67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-2AB**	**CFH30-2UUAB**	**CFH30-2RAB**	**CFH30-2UURAB**	**90**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CFH30-2VAB	**CFH30-2VUUAB**	**CFH30-2VRAB**	**CFH30-2VUURAB**													67 620	144 060	73 700	2 200				

CFH..AB TYPE has oil holes (grease nipple) which are prepared in the stud head and thread side, also an oil hole is prepared outer surface of stud.

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

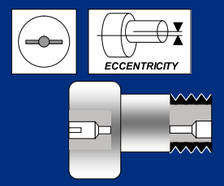
TYPE	Cylindrical outer ring	Crowned outer ring
CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 Grease Nipple	 NUT
All types	Installed	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
SCREWDRIVER SLOT HEAD

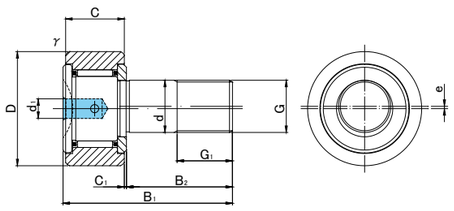




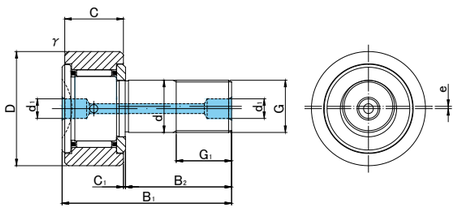
CFH



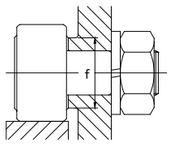
CFH..V



CFH5(V)～CFH10-1(V)



CFH12(V)～CFH16(V)



CFH TYPE

Prepacked Grease

Stud diameter (mm)		Designation				Dimensions (mm)											Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
		Cylindrical outer ring		Crowned outer ring R250(CF5) R500 (CF6 ～ CF16)																				
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	C1	d1	*r*s min	e	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
5	0 -0.012	**CFH 5**	**CFH 5UU**	**CFH 5R**	**CFH 5UUR**	**13**	9	5	M5×0.8	7.5	23	13	0.5	3.1	0.3	0.2	9.7	3 140	2 770	1 420	29 000	2 250	780	2	10.5
		CFH 5V	**CFH 5VUU**	**CFH 5VR**	**CFH 5VUUR**													5,100	5,500	1 420	11,600				
6	0 -0.012	**CFH 6**	**CFH 6UU**	**CFH 6R**	**CFH 6UUR**	**16**	11	6	M6×1	8	28	16	0.6	4	0.3	0.25	11	3 630	3 630	2 110	25 000	3 430	1 080	3	18.5
		CFH 6V	**CFH 6VUU**	**CFH 6VR**	**CFH 6VUUR**													6 960	8 530	2 110	12 000				
8	0 -0.015	**CFH 8**	**CFH 8UU**	**CFH 8R**	**CFH 8UUR**	**19**	11	8	M8×1.25	10	32	20	0.6	4	0.3	0.25	13	4 310	4 710	4 710	20 000	4 020	1 370	8	28.5
		CFH 8V	**CFH 8VUU**	**CFH 8VR**	**CFH 8VUUR**													8 130	11 170	4 710	9 000				
10	0 -0.015	**CFH10**	**CFH10UU**	**CFH10R**	**CFH10UUR**	**22**	12	10	M10×1.25	12	36	23	0.6	4	0.3	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CFH10V	**CFH10VUU**	**CFH10VR**	**CFH10VUUR**													9 510	14 500	7 450	7 500				
10	0 -0.015	**CFH10-1**	**CFH10-1UU**	**CFH10-1R**	**CFH10-1UUR**	**26**	12	10	M10×1.25	12	36	23	0.6	4	0.3	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CFH10-1V	**CFH10-1VUU**	**CFH10-1VR**	**CFH10-1VUUR**													9 510	14 500	7 450	7 500				
12	0 -0.018	**CFH12**	**CFH12UU**	**CFH12R**	**CFH12UUR**	**30**	14	12	M12×1.5	13	40	25	0.6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFH12V	**CFH12VUU**	**CFH12VR**	**CFH12VUUR**													13 430	19 700	11 270	6 000				
12	0 -0.018	**CFH12-1**	**CFH12-1UU**	**CFH12-1R**	**CFH12-1UUR**	**32**	14	12	M12×1.5	13	40	25	0.6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFH12-1V	**CFH12-1VUU**	**CFH12-1VR**	**CFH12-1VUUR**													13 430	19 700	11 270	6 000				
16	0 -0.018	**CFH16**	**CFH16UU**	**CFH16R**	**CFH16UUR**	**35**	18	16	M16×1.5	17	52	32.5	0.8	6	0.6	0.5	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFH16V	**CFH16VUU**	**CFH16VR**	**CFH16VUUR**													20 680	37 630	19 800	4 500				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

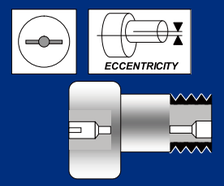
TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
SCREWDRIVER SLOT HEAD

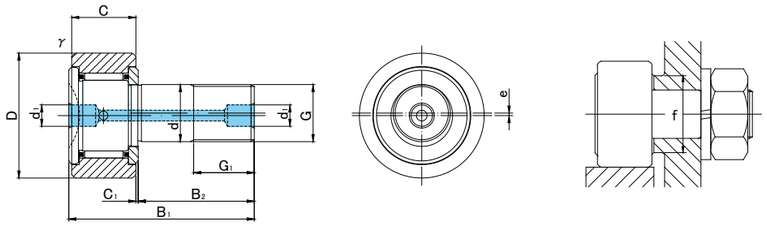




CFH



CFH..V



CFH TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R500(CF18) R1000(CF20 ≦), Without seals	Crowned outer ring, With seals	D	C	d	G	G1	B1	B2	C1	d1	r s min	e	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
18	0 -0.018	**CFH18**	**CFH18UU**	**CFH18R**	**CFH18UUR**	**40**	20	18	M18×1.5	19	58	36.5	0.8	6	1	0.6	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFH18V	**CFH18VUU**	**CFH18VR**	**CFH18VUUR**													25 280	51 350	26 560	3 500				
20	0 -0.021	**CFH20**	**CFH20UU**	**CFH20R**	**CFH20UUR**	**52**	24	20	M20×1.5	21	66	40.5	0.8	8	1	0.7	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFH20V	**CFH20VUU**	**CFH20VR**	**CFH20VUUR**													33 120	64 480	32 140	3 500				
20	0 -0.021	**CFH20-1**	**CFH20-1UU**	**CFH20-1R**	**CFH20-1UUR**	**47**	24	20	M20×1.5	21	66	40.5	0.8	8	1	0.7	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFH20-1V	**CFH20-1VUU**	**CFH20-1VR**	**CFH20-1VUUR**													33 120	64 480	32 140	3 500				
24	0 -0.021	**CFH24**	**CFH24UU**	**CFH24R**	**CFH24UUR**	**62**	29	24	M24×1.5	25	80	49.5	0.8	8	1	0.8	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFH24V	**CFH24VUU**	**CFH24VR**	**CFH24VUUR**													46 550	92 020	49 500	3 000				
24	0 -0.021	**CFH24-1**	**CFH24-1UU**	**CFH24-1R**	**CFH24-1UUR**	**72**	29	24	M24×1.5	25	80	49.5	0.8	8	1	0.8	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFH24-1V	**CFH24-1VUU**	**CFH24-1VR**	**CFH24-1VUUR**													46 550	92 020	49 500	3 000				
30	0 -0.021	**CFH30**	**CFH30UU**	**CFH30R**	**CFH30UUR**	**80**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFH30V	**CFH30VUU**	**CFH30VR**	**CFH30VUUR**													67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-1**	**CFH30-1UU**	**CFH30-1R**	**CFH30-1UUR**	**85**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CFH30-1V	**CFH30-1VUU**	**CFH30-1VR**	**CFH30-1VUUR**													67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-2**	**CFH30-2UU**	**CFH30-2R**	**CFH30-2UUR**	**90**	35	30	M30×1.5	32	100	63	1	8	1	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CFH30-2V	**CFH30-2VUU**	**CFH30-2VR**	**CFH30-2VUUR**													67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

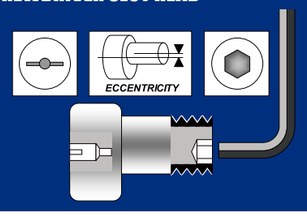
TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS

SOLID ECCENTRIC TYPE
HEXAGON SOCKET ON THREAD SIDE
SCREWDRIVER SLOT HEAD

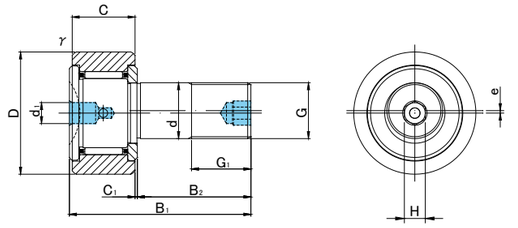




CFH..B



CFH..VB



CFH..B TYPE

Prepacked Grease

Stud diameter (mm)		Designation				Dimensions (mm)													Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
		Cylindrical outer ring		Crowned outer ring R500(CF12 ～ CF18) R1000(CF20 ≦)																						
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	C1	d1	H	*r*s min	e	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
12	0 -0.018	**CFH12B**	**CFH12UUB**	**CFH12RB**	**CFH12UURB**	**30**	14	12	M12×1.5	13	40	25	0.6	6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFH12VB	**CFH12VUUB**	**CFH12VRB**	**CFH12VUURB**														13 430	19 700	11 270	6 000				
12	0 -0.018	**CFH12-1B**	**CFH12-1UUB**	**CFH12-1RB**	**CFH12-1UURB**	**32**	14	12	M12×1.5	13	40	25	0.6	6	6	0.6	0.4	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFH12-1VB	**CFH12-1VUUB**	**CFH12-1VRB**	**CFH12-1VUURB**														13 430	19 700	11 270	6 000				
16	0 -0.018	**CFH16B**	**CFH16UUB**	**CFH16RB**	**CFH16UURB**	**35**	18	16	M16×1.5	17	52	32.5	0.8	6	6	0.6	0.5	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFH16VB	**CFH16VUUB**	**CFH16VRB**	**CFH16VUURB**														20 680	37 630	19 800	4 500				
18	0 -0.018	**CFH18B**	**CFH18UUB**	**CFH18RB**	**CFH18UURB**	**40**	20	18	M18×1.5	19	58	36.5	0.8	6	6	1	0.6	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFH18VB	**CFH18VUUB**	**CFH18VRB**	**CFH18VUURB**														25 280	51 350	26 560	3 500				
20	0 -0.021	**CFH20B**	**CFH20UUB**	**CFH20RB**	**CFH20UURB**	**52**	24	20	M20×1.5	21	66	40.5	0.8	8	8	1	0.7	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFH20VB	**CFH20VUUB**	**CFH20VRB**	**CFH20VUURB**														33 120	64 480	32 140	3 500				
20	0 -0.021	**CFH20-1B**	**CFH20-1UUB**	**CFH20-1RB**	**CFH20-1UURB**	**47**	24	20	M20×1.5	21	66	40.5	0.8	8	8	1	0.7	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFH20-1VB	**CFH20-1VUUB**	**CFH20-1VRB**	**CFH20-1VUURB**														33 120	64 480	32 140	3 500				
24	0 -0.021	**CFH24B**	**CFH24UUB**	**CFH24RB**	**CFH24UURB**	**62**	29	24	M24×1.5	25	80	49.5	0.8	8	8	1	0.8	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFH24VB	**CFH24VUUB**	**CFH24VRB**	**CFH24VUURB**														46 550	92 020	49 500	3 000				
24	0 -0.021	**CFH24-1B**	**CFH24-1UUB**	**CFH24-1RB**	**CFH24-1UURB**	**72**	29	24	M24×1.5	25	80	49.5	0.8	8	8	1	0.8	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFH24-1VB	**CFH24-1VUUB**	**CFH24-1VRB**	**CFH24-1VUURB**														46 550	92 020	49 500	3 000				
30	0 -0.021	**CFH30B**	**CFH30UUB**	**CFH30RB**	**CFH30UURB**	**80**	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFH30VB	**CFH30VUUB**	**CFH30VRB**	**CFH30VUURB**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-1B**	**CFH30-1UUB**	**CFH30-1RB**	**CFH30-1UURB**	**85**	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CFH30-1VB	**CFH30-1VUUB**	**CFH30-1VRB**	**CFH30-1VUURB**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CFH30-2B**	**CFH30-2UUB**	**CFH30-2RB**	**CFH30-2UURB**	**90**	35	30	M30×1.5	32	100	63	1	8	8	1	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CFH30-2VB	**CFH30-2VUUB**	**CFH30-2VRB**	**CFH30-2VUURB**														67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

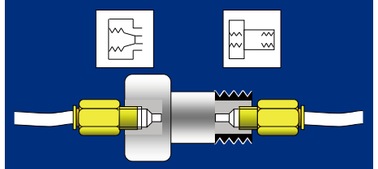
TYPE	Cylindrical outer ring	Crowned outer ring
CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS

TAP HOLE FOR PIPING
SCREWDRIVER SLOT HEAD

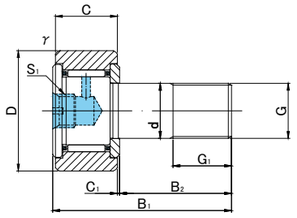




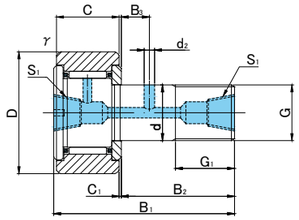
CFT



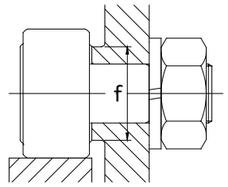
CFT..V



CFT6(V)～CFT10-1(V)



CFT12(V)～CFT18(V)



CFT TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Designation: Cylindrical outer ring, With seals	Designation: Crowned outer ring R500(CF6 ～ CF18), Without seals	Designation: Crowned outer ring R500(CF6 ～ CF18), With seals	D	C	d	G	G1	B1	B2	B3	C1	S1	d2	r's min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
6	0 -0.012	**CFT 6**	**CFT 6UU**	**CFT 6R**	**CFT 6UUR**	**16**	11	6	M6×1	8	28	16	—	0.6	M6×0.75	—	0.3	11	3 630	3 630	2 110	25 000	3 430	1 080	3	18.5
		CFT 6V	**CFT 6VUU**	**CFT 6VR**	**CFT 6VUUR**														6 960	8 530	2 110	12 000				
8	0 -0.015	**CFT 8**	**CFT 8UU**	**CFT 8R**	**CFT 8UUR**	**19**	11	8	M8×1.25	10	32	20	—	0.6	M6×0.75	—	0.3	13	4 310	4 710	4 710	20 000	4 020	1 370	8	28.5
		CFT 8V	**CFT 8VUU**	**CFT 8VR**	**CFT 8VUUR**														8 130	11 170	4 710	9 000				
10	0 -0.015	**CFT10**	**CFT10UU**	**CFT10R**	**CFT10UUR**	**22**	12	10	M10×1.25	12	36	23	—	0.6	M6×0.75	—	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CFT10V	**CFT10VUU**	**CFT10VR**	**CFT10VUUR**														9 510	14 500	7 450	7 500				
10	0 -0.015	**CFT10-1**	**CFT10-1UU**	**CFT10-1R**	**CFT10-1UUR**	**26**	12	10	M10×1.25	12	36	23	—	0.6	M6×0.75	—	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CFT10-1V	**CFT10-1VUU**	**CFT10-1VR**	**CFT10-1VUUR**														9 510	14 500	7 450	7 500				
12	0 -0.018	**CFT12**	**CFT12UU**	**CFT12R**	**CFT12UUR**	**30**	14	12	M12×1.5	13	40	25	6	0.6	M6×0.75	3	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFT12V	**CFT12VUU**	**CFT12VR**	**CFT12VUUR**														13 430	19 700	11 270	6 000				
12	0 -0.018	**CFT12-1**	**CFT12-1UU**	**CFT12-1R**	**CFT12-1UUR**	**32**	14	12	M12×1.5	13	40	25	6	0.6	M6×0.75	3	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFT12-1V	**CFT12-1VUU**	**CFT12-1VR**	**CFT12-1VUUR**														13 430	19 700	11 270	6 000				
16	0 -0.018	**CFT16**	**CFT16UU**	**CFT16R**	**CFT16UUR**	**35**	18	16	M16×1.5	17	52	32.5	8	0.8	Rc1/8	3	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFT16V	**CFT16VUU**	**CFT16VR**	**CFT16VUUR**														20 680	37 630	19 800	4 500				
18	0 -0.018	**CFT18**	**CFT18UU**	**CFT18R**	**CFT18UUR**	**40**	20	18	M18×1.5	19	58	36.5	8	0.8	Rc1/8	3	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFT18V	**CFT18VUU**	**CFT18VR**	**CFT18VUUR**														25 280	51 350	26 560	3 500				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

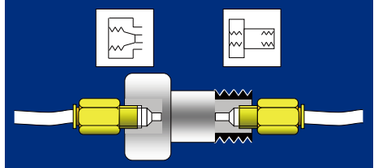
TYPE	Cylindrical outer ring	Crowned outer ring
CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 NUT
All types	Attached

CAM FOLLOWERS

TAP HOLE FOR PIPING
SCREWDRIVER SLOT HEAD

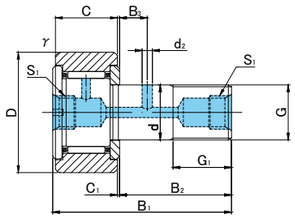


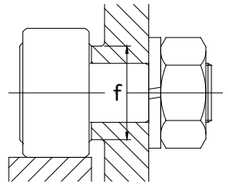


CFT



CFT..V





CFT TYPE

Prepacked Grease

Stud diameter (mm)		Designation				Dimensions (mm)												Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
		Cylindrical outer ring		Crowned outer ring R1000(CF20 ≦)																					
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	B3	C1	S1	d2	r s min	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
20	0 -0.021	**CFT20**	**CFT20UU**	**CFT20R**	**CFT20UUR**	**52**	24	20	M20×1.5	21	66	40.5	9	0.8	Rc1/8	4	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFT20V	**CFT20VUU**	**CFT20VR**	**CFT20VUUR**														33 120	64 480	32 140	3 500				
20	0 -0.021	**CFT20-1**	**CFT20-1UU**	**CFT20-1R**	**CFT20-1UUR**	**47**	24	20	M20×1.5	21	66	40.5	9	0.8	Rc1/8	4	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFT20-1V	**CFT20-1VUU**	**CFT20-1VR**	**CFT20-1VUUR**														33 120	64 480	32 140	3 500				
24	0 -0.021	**CFT24**	**CFT24UU**	**CFT24R**	**CFT24UUR**	**62**	29	24	M24×1.5	25	80	49.5	11	0.8	Rc1/8	4	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFT24V	**CFT24VUU**	**CFT24VR**	**CFT24VUUR**														46 550	92 020	49 500	3 000				
24	0 -0.021	**CFT24-1**	**CFT24-1UU**	**CFT24-1R**	**CFT24-1UUR**	**72**	29	24	M24×1.5	25	80	49.5	11	0.8	Rc1/8	4	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFT24-1V	**CFT24-1VUU**	**CFT24-1VR**	**CFT24-1VUUR**														46 550	92 020	49 500	3 000				
30	0 -0.021	**CFT30**	**CFT30UU**	**CFT30R**	**CFT30UUR**	**80**	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFT30V	**CFT30VUU**	**CFT30VR**	**CFT30VUUR**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CFT30-1**	**CFT30-1UU**	**CFT30-1R**	**CFT30-1UUR**	**85**	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	1	46	45 370	85 060	73 700	5 000	56 000	56 000	450	2 030
		CFT30-1V	**CFT30-1VUU**	**CFT30-1VR**	**CFT30-1VUUR**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CFT30-2**	**CFT30-2UU**	**CFT30-2R**	**CFT30-2UUR**	**90**	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	1	46	45 370	85 060	73 700	5 000	59 300	59 300	450	2 220
		CFT30-2V	**CFT30-2VUU**	**CFT30-2VR**	**CFT30-2VUUR**														67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

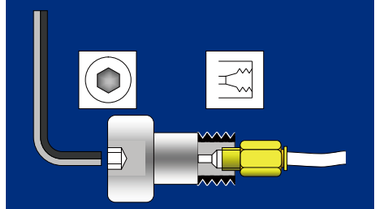
TYPE	Cylindrical outer ring	Crowned outer ring
CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 NUT
All types	Attached

CAM FOLLOWERS

TAP HOLE FOR PIPING
HEXAGON SOCKET ON STUD HEAD

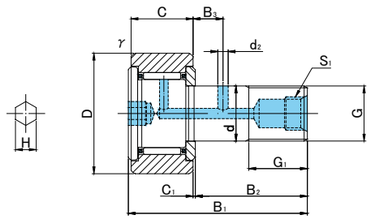


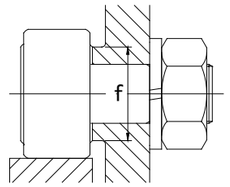


CFT..A



CFT..VA





CFT..A TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R500(CF12～CF18) R1000(CF20≦), Without seals	Crowned outer ring, With seals	D	C	d	G	G1	B1	B2	B3	C1	S1	d2	H	r's min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
12	0 -0.018	CFT12A	CFT12UUA	CFT12RA	CFT12UURA	30	14	12	M12×1.5	13	40	25	6	0.6	M6×0.75	3	6	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CFT12VA	CFT12VUUA	CFT12VRA	CFT12VUURA															13 430	19 700	11 270	6 000				
12	0 -0.018	CFT12-1A	CFT12-1UUA	CFT12-1RA	CFT12-1UURA	32	14	12	M12×1.5	13	40	25	6	0.6	M6×0.75	3	6	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CFT12-1VA	CFT12-1VUUA	CFT12-1VRA	CFT12-1VUURA															13 430	19 700	11 270	6 000				
16	0 -0.018	CFT16A	CFT16UUA	CFT16RA	CFT16UURA	35	18	16	M16×1.5	17	52	32.5	8	0.8	Rc1/8	3	6	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CFT16VA	CFT16VUUA	CFT16VRA	CFT16VUURA															20 680	37 630	19 800	4 500				
18	0 -0.018	CFT18A	CFT18UUA	CFT18RA	CFT18UURA	40	20	18	M18×1.5	19	58	36.5	8	0.8	Rc1/8	3	6	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CFT18VA	CFT18VUUA	CFT18VRA	CFT18VUURA															25 280	51 350	26 560	3 500				
20	0 -0.021	CFT20A	CFT20UUA	CFT20RA	CFT20UURA	52	24	20	M20×1.5	21	66	40.5	9	0.8	Rc1/8	4	8	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CFT20VA	CFT20VUUA	CFT20VRA	CFT20VUURA															33 120	64 480	32 140	3 500				
20	0 -0.021	CFT20-1A	CFT20-1UUA	CFT20-1RA	CFT20-1UURA	47	24	20	M20×1.5	21	66	40.5	9	0.8	Rc1/8	4	8	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CFT20-1VA	CFT20-1VUUA	CFT20-1VRA	CFT20-1VUURA															33 120	64 480	32 140	3 500				
24	0 -0.021	CFT24A	CFT24UUA	CFT24RA	CFT24UURA	62	29	24	M24×1.5	25	80	49.5	11	0.8	Rc1/8	4	8	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CFT24VA	CFT24VUUA	CFT24VRA	CFT24VUURA															46 550	92 020	49 500	3 000				
24	0 -0.021	CFT24-1A	CFT24-1UUA	CFT24-1RA	CFT24-1UURA	72	29	24	M24×1.5	25	80	49.5	11	0.8	Rc1/8	4	8	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CFT24-1VA	CFT24-1VUUA	CFT24-1VRA	CFT24-1VUURA															46 550	92 020	49 500	3 000				
30	0 -0.021	CFT30A	CFT30UUA	CFT30RA	CFT30UURA	80	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	8	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CFT30VA	CFT30VUUA	CFT30VRA	CFT30VUURA															67 620	144 060	73 700	2 200				
30	0 -0.021	CFT30-1A	CFT30-1UUA	CFT30-1RA	CFT30-1UURA	85	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	8	1	46	45 370	85 060	73 700	5 000	56 000	56 000	450	2 030
		CFT30-1VA	CFT30-1VUUA	CFT30-1VRA	CFT30-1VUURA															67 620	144 060	73 700	2 200				
30	0 -0.021	CFT30-2A	CFT30-2UUA	CFT30-2RA	CFT30-2UURA	90	35	30	M30×1.5	32	100	63	15	1	Rc1/8	4	8	1	46	45 370	85 060	73 700	5 000	59 300	59 300	450	2 220
		CFT30-2VA	CFT30-2VUUA	CFT30-2VRA	CFT30-2VUURA															67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE

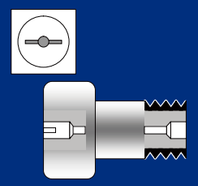
TYPE	Cylindrical outer ring	Crowned outer ring
CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 NUT
All types	Attached

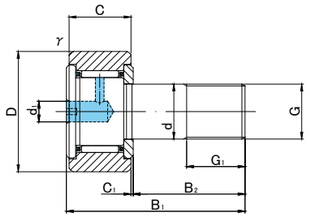
CAM FOLLOWERS

INCH DIMENSION
SCREWDRIVER SLOT HEAD

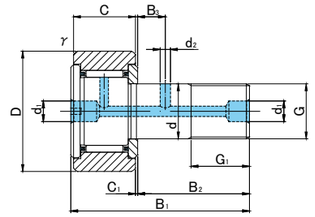




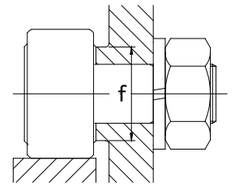
CR



CR8～CR10-1



CR12～CR36



CR TYPE

Prepacked Grease

Stud diameter d mm (inch)	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R250(≦CR8-1) R500(CR10≦), Without seals	Crowned outer ring R250(≦CR8-1) R500(CR10≦), With seals	Dimensions (inch/mm): D		C		d		G UNF	G1		B1		B2		B3		C1	d1		d2		r		f		Basic dynamic load rating Cr N	Basic static load rating Cor N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
4.826 (-)	CR8	CR8UU	CR8R	CR8UUR	1/2	12.7	11/32	8.731	—	4.826	No.10-32	1/4	6.35	7/8	22.225	1/2	12.7	—	—	0.8	1/8	3.175	—	—	1/64	0.397	21/64	8.334	2 550	2 160	28 000	2 160	730	2	9
	CR8-1	CR8-1UU	CR8-1R	CR8-1UUR			3/8	9.525						1 1/32	26.194	5/8	15.875															2 350	730		10
6.350 (1/4)	CR10	CR10UU	CR10R	CR10UUR	5/8	15.875	13/32	10.319	1/4	6.35	1/4 - 28	5/16	7.938	1 1/16	26.988	5/8	15.875	—	—	0.8	1/8	3.175	—	—	1/64	0.397	29/64	11.509	3 630	3 630	21 000	3 230	1 180	3	19
	CR10-1	CR10-1UU	CR10-1R	CR10-1UUR			7/16	11.112						1 7/32	30.956	3/4	19.05															3 430	1 180		21
9.525 (3/8)	CR12	CR12UU	CR12R	CR12UUR	3/4	19.05	1/2	12.7	3/8	9.525	3/8 - 24	3/8	9.525	1 13/32	35.719	7/8	22.225	1/4	6.35	0.8	3/16	4.762	3/32	2.381	1/32	0.794	17/32	13.494	4 410	5 100	15 000	4 510	1 270	17	35
	CR14	CR14UU	CR14R	CR14UUR	7/8	22.225																				19/32	15.081	4 800	5 780	14 000	5 290	1 760		46
11.112 (7/16)	CR16	CR16UU	CR16R	CR16UUR	1	25.4	5/8	15.875	7/16	11.112	7/16 - 20	1/2	12.7	1 21/32	42.069	1	25.4	1/4	6.35	0.8	3/16	4.762	1/8	3.175	3/64	1.191	45/64	17.859	8 820	10 780	13 000	7 250	1 960	20	73
	CR18	CR18UU	CR18R	CR18UUR	1 1/8	28.575																			1/16	1.588	3/4	19.05	9 210	11 560	12 000	7 740	2 350		88
12.700 (1/2)	CR20	CR20UU	CR20R	CR20UUR	1 1/4	31.75	3/4	19.05	1/2	12.7	1/2 - 20	5/8	15.875	2 1/32	51.594	1 1/4	31.75	5/16	7.938	0.8	3/16	4.762	1/8	3.175	1/16	1.588	55/64	21.828	14 210	15 970	11 000	10 680	2 840	28	132
	CR22	CR22UU	CR22R	CR22UUR	1 3/8	34.925																										11 760	2 940		157
15.875 (5/8)	CR24	CR24UU	CR24R	CR24UUR	1 1/2	38.1	7/8	22.225	5/8	15.875	5/8 - 18	3/4	19.05	2 13/32	61.119	1 1/2	38.1	3/8	9.525	0.8	3/16	4.762	5/32	3.969	1/16	1.588	1 3/64	26.196	18 520	24 210	8 500	15 390	3 820	64	225
	CR26	CR26UU	CR26R	CR26UUR	1 5/8	41.275																										16 660	3 820		260
19.05 (3/4)	CR28	CR28UU	CR28R	CR28UUR	1 3/4	44.45	1	25.4	3/4	19.05	3/4 - 16	7/8	22.225	2 25/32	70.644	1 3/4	44.45	7/16	11.112	0.8	3/16	4.762	5/32	3.969	1/16	1.588	1 9/32	32.543	25 090	38 220	7 000	20 970	3 820	117	365
	CR30	CR30UU	CR30R	CR30UUR	1 7/8	47.625																										22 440	4 700		410
22.225 (7/8)	CR32	CR32UU	CR32R	CR32UUR	2	50.8	1 1/4	31.75	7/8	22.225	7/8 - 14	1	25.4	3 9/32	83.344	2	50.8	1/2	12.7	0.8	3/16	4.762	3/16	4.762	1/16	1.588	1 15/32	37.306	32 440	63 800	5 500	30 870	4 900	186	615
	CR36	CR36UU	CR36R	CR36UUR	2 1/4	57.15																										34 690	5 980		750

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

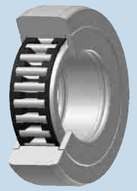
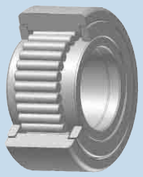
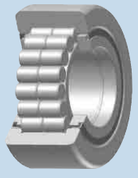
ACCESSORIES

TYPE	STOP PLUG	RESIN PLUG	NUT
CR8, CR8-1,CR10,CR10-1	—	—	Attached
CR12.CR14,CR16,CR18,CR20,CR22	ϕ 4.8 Attached	ϕ 4.8 Attached	Attached
CR24,CR26,CR28,CR30,CR32,CR36	ϕ 4.8 Attached	ϕ 4.8 Attached	Attached

ROLLER FOLLOWERS



Type and Part Code

Type	Applicable shaft diameter	Feature	Part Code
 RNAST (Separable type)	ϕ 7 ~ ϕ 60	NAST type without inner ring. Available with stainless steel type (code M) for higher corrosive resistance.	**RNAST** **15** **M** **R** Type **M**: Stainless steel **None**: High Carbon steel **R**: Crowned outer ring **None**: Cylindrical outer ring
 NAST (Separable type)	ϕ 6 ~ ϕ 50	Thick wall outer ring, inner ring. Separable bearing with combined needle roller with precision cage. Available with stainless steel type (code M) for higher corrosive resistance.	**NAST** **15** **M** **R** Type **M**: Stainless steel **None**: High Carbon steel **R**: Crowned outer ring **None**: Cylindrical outer ring
 NAST-ZZ (Separable type)	ϕ 6 ~ ϕ 50	Separable bearing in which labyrinth seal is formed with combined side plate at both sides of inner ring of NAST type. (NAST-ZZUU type comes with seal) Available with stainless steel type (code M) for higher corrosive resistance.	**NAST** **15** **M** **Z Z** **U U** **R** Type **M**: Stainless steel **None**: High Carbon steel **Z Z**: With shield **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring
 NART-R (Non-separable type)	ϕ 5 ~ ϕ 50	Non-separable bearing with fixed side plate at inner ring. Mitigate eccentric load with spherical shape at outer diameter of outer ring (Code R). (NART-UUR type comes with seal) Available with stainless steel type (code M) for higher corrosive resistance.	**NART** **15** **M** **U U** **V** **R** Type **M**: Stainless steel **None**: High Carbon steel **UU**: With seal **None**: With shield **V**: Full complement **None**: With cage **R**: Crowned outer ring **X**: Cylindrical outer ring
 NURT-R (Double-row cylindrical roller)	ϕ 15 ~ ϕ 50	Bearing which allows high radial loading and moderate axial loading with integrated cylindrical roller in double rows. Mitigate eccentric load with spherical shape at outer diameter of outer ring (Code R).	**NURT** **15** **R** Type **R**: Crowned outer ring **X**: Cylindrical outer ring

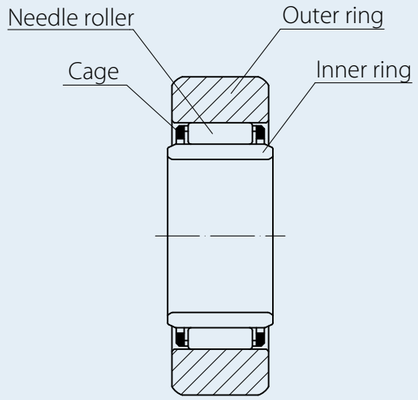
Structure and Features

Roller follower is a bearing with integrated “roller” featuring high rigidity. Its primary application is in a guide roller to have its contacting member travel in linier motion in direction of tangent using rotation of outer ring.

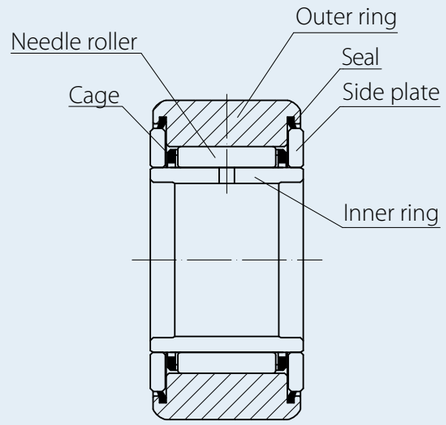
Outer ring of the roller follower is designed with thick ring in order to provide high rigidity to endure heavy load or impact load at portion to contact with the associated objects.

Type of roller follower mainly consists of separate type, which have separated inner ring and outer ring, and non-separate type, whose inner ring has fixed side plate.

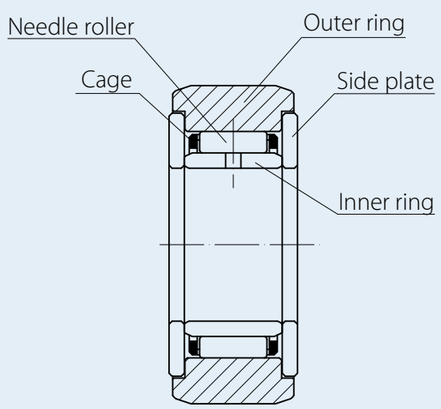
Also, various types with shapes and structures are available in accordance with application. Outer rings have two types – crowned outer ring and cylindrical outer ring. The crowned outer ring is suited for absorbing eccentric load since it absorbs angular misalignment due to a mount error. Cylindrical outer ring is effective for heavy duty since it contacts with large contact area of the associated objects so that can mitigate the contact surface pressure. They have two internal mechanism – one with cage and one with full complement. A roller follower with cage is suitable for high speed application since the guiding feature of cage enables the rotation of "rollers" to be stable. Compared with cage type, full complement type is effective for applications with low-speed operation of heavy load due to its larger load rating. Full complement type also includes double-raw cylindrical roller type which allows moderate axial loading.



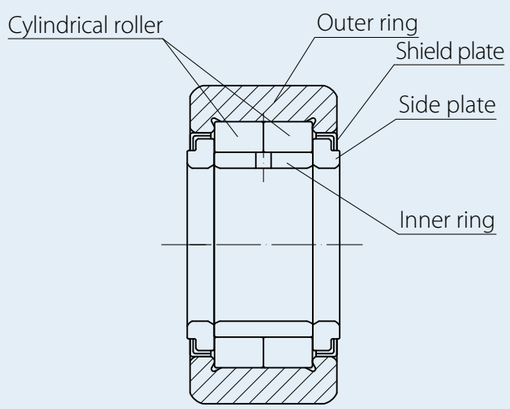
Structure of type NAST



Structure of type NART..UUR



Structure of type NAST..ZZ



Structure of type NURT

Accuracy standard

Roller followers are manufactured in accordance with the following accuracies.

Table-1 Accuracy Unit: µm

Name \ Category		Crowned outer ring	Cylindrical outer ring
Dimension tolerance of outer ring outer diameter (D)		0 -50	Refer to Table-3
Dimension tolerance of outer ring width (C)		0 -120	
Tolerance of inner ring width (B)	Separable	0 -120	
Tolerance of bearing width (B)	Non-separable	h12	—
Fw	Separable	Refer to Table-11 on page 23	

Table-2 Accuracy of inner ring Unit: µm

Nominal inner ring bore diameter d (mm)		Deviation of mean bore diameter in a single plane Δ_{dmp}		Tolerance of radial runout of inner ring
Over	Incl.	high	low	max.
2.5	10	0	-8	10
10	18	0	-9	10
18	30	0	-10	13
30	50	0	-12	15

Table-3 Accuracy of outer ring Unit: µm

Nominal outer ring bore diameter D (mm)		Deviation of mean outer diameter in a single plane Δ_{Dmp}		Tolerance of radial runout of outer ring (max.)
Over	Incl.	high	low	max.
6	18	0	-8	15
18	30	0	-9	15
30	50	0	-11	20
50	80	0	-13	25
80	120	0	-15	35

Radial internal clearance

Table below indicates radial internal clearance of roller follower.

Table-4 Radial internal clearance

Unit: μm

Part code			Radial internal clearance	
Separable	Non-separable	Double row cylindrical roller	min.	max.
NAST6	NART5R ～ 6R		5	20
NAST8 ～ 12	NART8R ～ 12R		5	25
NAST15 ～ 25	NART15R ～ 25R		10	30
NAST30 ～ 40	NART30R ～ 40R		10	40
NAST45 ～ 50	NART45R ～ 50R		15	50
		NURT15R ～ 30-1R	0	25
		NURT35R ～ 40-1R	5	30
		NURT45R ～ 50-1R	5	35

Fits

Table below indicates recommended fits between roller follower and its mounting shaft.

Table-5 Shaft fits

Type		Tolerance grade
Separable	Without inner ring	k5,k6
	With inner ring	g6,h6
Non-separable		
Double row cylindrical roller		

Track load capacity

Track load capacity means a permissive load under which the outer ring of cam follower and the mating surface are allowable to be used over a long period without causing any deformation nor compression mark. Track load capacity depicted in dimension table indicates a value for which hardness of contacting steel member is assumed to be HRC40. In the case that the hardness of contacting member is not HRC40, track load capacity in the dimension table shall be multiplied by value of track capacity factor obtained by Figure-1.

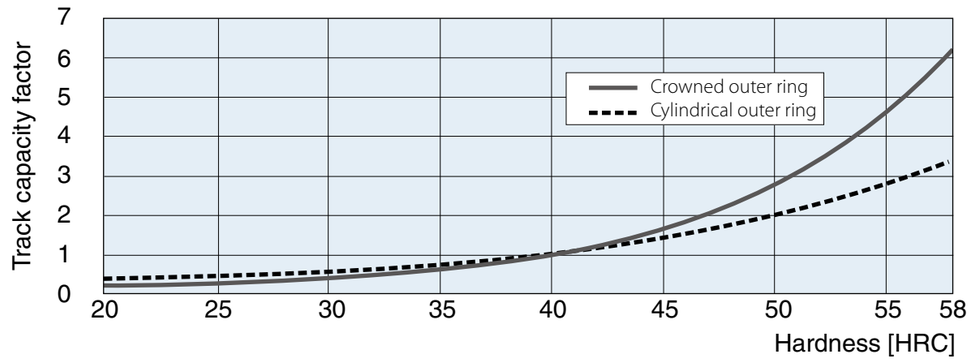


Figure-1 Track capacity factor

Lubrication

All the JNS roller followers are lubricated with pre-packed high quality lithium soap-based grease grade 2 (RoHS compliant) so that they are ready to use. For applications requiring high prevention measure against invasion of foreign matter or leakage of lubricant, products with seal (part code--UU) which integrate special synthetic lubber with high abrasion resistance are also available.

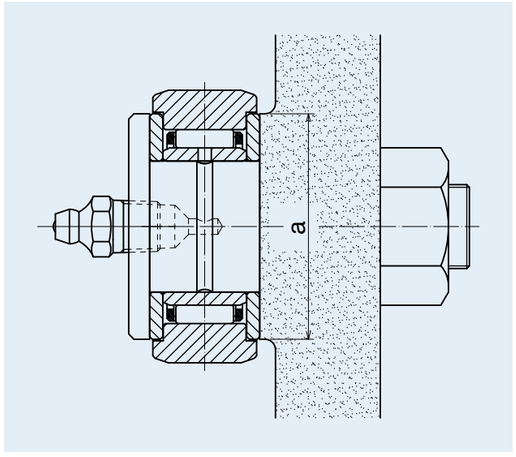
Grease shall be packed up to volume approximately one-third to one-half of internal space of bearing. Lubrication interval varies depending on operation condition. Referential interval may be every six to twenty four months for roller followers with cages and every one to six months for full complement type with grease in the same type.

Some excessive grease may leak at the beginning of usage or immediately after re-greasing even for the products with seal. Aging operation period is recommended prior to application in which no contamination by grease is allowed in and around device. Wipe and clean any leaked extra grease after this operation.

Mounting

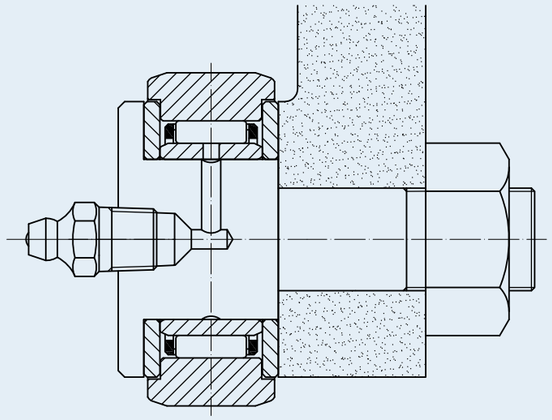
Mounting part

- Thrust load may cause damage to side plate, outer ring or roller follower since it is designed so as to support load in radial direction. It is recommended to design and assemble to avoid thrust load.
- Application with roller follower without inner ring requires heat treatment and grinding finish of shaft. Surface hardness of the shaft shall be HRC58 to 64, and surface roughness shall be R_a 0.2 μ m or less for this application.
- Side plate of non-separate type roller follower is pressed-in to fix it. Application in a manner to push the side plate shall be avoided in order to prevent risk of abnormal rotation under external force.
- Crowned outer ring is recommended in the case that contact between the outer ring and contacting member track surface is not smooth and even.
- When mounting NART, NAST-ZZ and NURT type, dimension "a" must be more than it described in dimension table in order to protect the side plate.

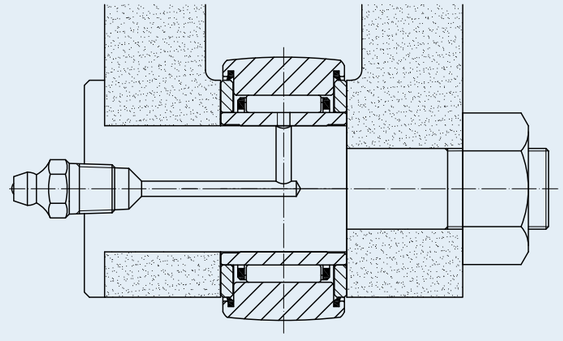


Mounting method

To prevent pre-mature failure of roller follower, lubrication hole of inner ring shall be located outside of loading range (on the side to receive load).



NAST..ZZ Example of mounting



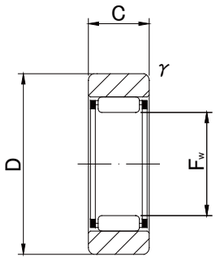
NART..UUR Example of mounting

ROLLER FOLLOWERS
SEPARABLE WITHOUT INNER RING

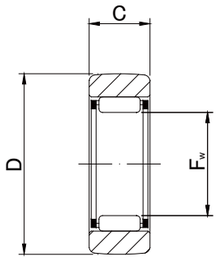




RNAST



RNAST



RNAST..R

RNAST TYPE

Prepacked Grease

Shaft Diameter (mm)	Designation		Dimensions (mm)				Basic dynamic load rating	Basic static load rating	Track load capacity		Limiting speed *	Mass
	Cylindrical outer ring	Crowned outer ring (R500)	Fw	D	C	r s min	Cr N	Cor N	Cylindrical outer ring N	Crowned outer ring N	rpm	g (approx)
7	**RNAST 5**	**RNAST 5R**	7 +0.022 +0.013	**16**	7.8	0.3	2 740	2 390	2 350	1 080	30 000	8.9
10	**RNAST 6**	**RNAST 6R**	10 +0.027 +0.016	**19**	9.8	0.3	4 120	4 550	3 530	1 370	20 000	13.9
12	**RNAST 8**	**RNAST 8R**	12 +0.027 +0.016	**24**	9.8	0.6	5 680	5 890	4 020	1 860	17 000	23.5
14	**RNAST10**	**RNAST10R**	14 +0.027 +0.016	**30**	11.8	1	9 700	9 670	5 590	2 450	15 000	42.5
16	**RNAST12**	**RNAST12R**	16 +0.027 +0.016	**32**	11.8	1	10 400	10 900	5 980	2 740	13 000	49.5
20	**RNAST15**	**RNAST15R**	20 +0.033 +0.020	**35**	11.8	1	12 300	14 300	6 570	3 140	10 000	50
22	**RNAST17**	**RNAST17R**	22 +0.033 +0.020	**40**	15.8	1	17 400	20 900	10 900	3 720	9 500	90
25	**RNAST20**	**RNAST20R**	25 +0.033 +0.020	**47**	15.8	1	19 200	24 500	12 700	4 610	8 500	135
30	**RNAST25**	**RNAST25R**	30 +0.033 +0.020	**52**	15.8	1	20 700	28 400	14 100	5 290	7 000	152
38	**RNAST30**	**RNAST30R**	38 +0.041 +0.025	**62**	19.8	1	30 300	45 400	22 100	6 660	5 500	255
42	**RNAST35**	**RNAST35R**	42 +0.041 +0.025	**72**	19.8	1	32 200	50 600	25 700	8 130	5 000	375
50	**RNAST40**	**RNAST40R**	50 +0.041 +0.025	**80**	19.8	1.5	35 700	61 100	26 900	9 310	4 000	420
55	**RNAST45**	**RNAST45R**	55 +0.049 +0.030	**85**	19.8	1.5	37 100	66 400	28 500	10 100	4 000	460
60	**RNAST50**	**RNAST50R**	60 +0.049 +0.030	**90**	19.8	1.5	38 700	71 800	30 200	11 000	3 500	500

* Suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible.

OUTER RINGS TOLERANCE (μm)

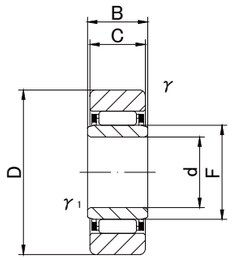
TYPE	Cylindrical outer ring	Crowned outer ring
RNAST5	0/-8	0/-50
RNAST6,RNAST8,RNAST10	0/-9	0/-50
RNAST12,RNAST15,RNAST17,RNAST20	0/-11	0/-50
RNAST25,RNAST30,RNAST35,RNAST40	0/-13	0/-50
RNAST45,RNAST50	0/-15	0/-50

ROLLER FOLLOWERS
SEPARABLE WITH INNER RING

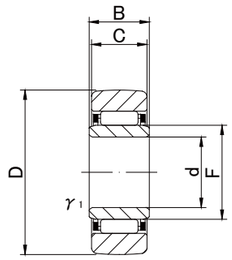




NAST



NAST



NAST..R

NAST TYPE

Prepacked Grease

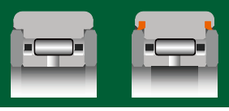
Shaft Diameter (mm)	Designation		Dimensions (mm)							Basic dynamic load rating	Basic static load rating	Track load capacity		Limiting speed *	Mass
	Cylindrical outer ring	Crowned outer ring (R500)	d	D	B	C	r s min	r 1s min	F	Cr N	Cor N	Cylindrical outer ring N	Crowned outer ring N	rpm	g (approx)
6	**NAST 6**	**NAST 6R**	6 0 -0.008	**19**	10	9.8	0.3	0.3	10	4 120	4 550	3 530	1 370	20 000	17.8
8	**NAST 8**	**NAST 8R**	8 0 -0.008	**24**	10	9.8	0.6	0.3	12	5 680	5 890	4 020	1 860	17 000	28
10	**NAST10**	**NAST10R**	10 0 -0.008	**30**	12	11.8	1	0.3	14	9 700	9 670	5 590	2 450	15 000	50
12	**NAST12**	**NAST12R**	12 0 -0.008	**32**	12	11.8	1	0.3	16	10 400	10 900	5 980	2 740	13 000	58
15	**NAST15**	**NAST15R**	15 0 -0.008	**35**	12	11.8	1	0.3	20	12 300	14 300	6 570	3 140	10 000	62
17	**NAST17**	**NAST17R**	17 0 -0.010	**40**	16	15.8	1	0.3	22	17 400	20 900	10 900	3 720	9 500	110
20	**NAST20**	**NAST20R**	20 0 -0.010	**47**	16	15.8	1	0.3	25	19 200	24 500	12 700	4 610	8 500	155
25	**NAST25**	**NAST25R**	25 0 -0.010	**52**	16	15.8	1	0.3	30	20 700	28 400	14 100	5 290	7 000	180
30	**NAST30**	**NAST30R**	30 0 -0.010	**62**	20	19.8	1	0.6	38	30 300	45 400	22 100	6 660	5 500	320
35	**NAST35**	**NAST35R**	35 0 -0.012	**72**	20	19.8	1	0.6	42	32 200	50 600	25 700	8 130	5 000	440
40	**NAST40**	**NAST40R**	40 0 -0.012	**80**	20	19.8	1.5	1	50	35 700	61 100	26 900	9 310	4 000	530
45	**NAST45**	**NAST45R**	45 0 -0.012	**85**	20	19.8	1.5	1	55	37 100	66 400	28 500	10 100	4 000	580
50	**NAST50**	**NAST50R**	50 0 -0.012	**90**	20	19.8	1.5	1	60	38 700	71 800	30 200	11 000	3 500	635

* Suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible.

OUTER RINGS TOLERANCE (Outside diameter) (μm)

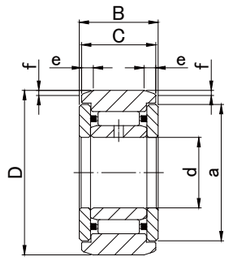
TYPE	Cylindrical outer ring	Crowned outer ring
NAST6,NAST8,NAST10	0/-9	0/-50
NAST12,NAST15,NAST17,NAST20	0/-11	0/-50
NAST25,NAST30,NAST35,NAST40	0/-13	0/-50
NAST45,NAST50	0/-15	0/-50

ROLLER FOLLOWERS
SEPARABLE
WITH INNER RING
WITH SHIELD

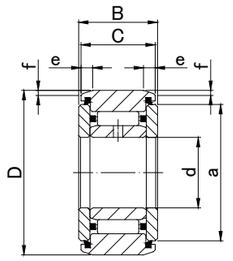




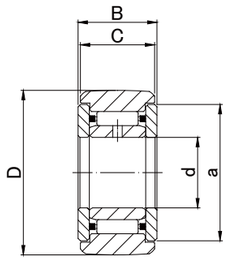
NAST..ZZ



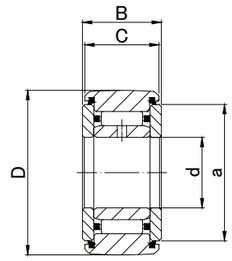
NAST..ZZ



NAST..ZZUU



NAST..ZZR



NAST..ZZUUR

NAST..ZZ TYPE

Prepacked Grease

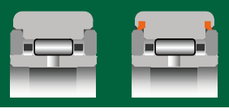
Shaft Diameter (mm)	Designation: Cylindrical outer ring, Without seals	Designation: Cylindrical outer ring, With seals	Designation: Crowned outer ring (R500), Without seals	Designation: Crowned outer ring (R500), With seals	Dimensions (mm): d	D	B	C	a	e	f	rs min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Limiting speed * rpm	Mass g (approx)
6	NAST 6ZZ	NAST 6ZZUU	NAST 6ZZR	NAST 6ZZUUR	6 0 -0.008	19	14	13.8	14	2.5	0.8	0.3	4 120	4 550	3 530	1 370	20 000	24.5
8	NAST 8ZZ	NAST 8ZZUU	NAST 8ZZR	NAST 8ZZUUR	8 0 -0.008	24	14	13.8	17.5	2.5	0.8	0.6	5 680	5 890	4 510	1 860	17 000	39
10	NAST10ZZ	NAST10ZZUU	NAST10ZZR	NAST10ZZUUR	10 0 -0.008	30	16	15.8	23.5	2.5	0.8	1	9 700	9 670	6 860	2 450	15 000	65
12	NAST12ZZ	NAST12ZZUU	NAST12ZZR	NAST12ZZUUR	12 0 -0.008	32	16	15.8	25.5	2.5	0.8	1	10 400	10 900	7 350	2 740	13 000	75
15	NAST15ZZ	NAST15ZZUU	NAST15ZZR	NAST15ZZUUR	15 0 -0.008	35	16	15.8	29	2.5	0.8	1	12 300	14 300	8 040	3 140	10 000	83
17	NAST17ZZ	NAST17ZZUU	NAST17ZZR	NAST17ZZUUR	17 0 -0.010	40	20	19.8	32.5	3	1	1	17 400	20 900	11 800	3 720	9 500	135
20	NAST20ZZ	NAST20ZZUU	NAST20ZZR	NAST20ZZUUR	20 0 -0.010	47	20	19.8	38	3	1	1	19 200	24 500	13 800	4 610	8 500	195
25	NAST25ZZ	NAST25ZZUU	NAST25ZZR	NAST25ZZUUR	25 0 -0.010	52	20	19.8	43	3	1	1	20 700	28 400	15 300	5 290	7 000	225
30	NAST30ZZ	NAST30ZZUU	NAST30ZZR	NAST30ZZUUR	30 0 -0.010	62	25	24.8	50.5	4	1.2	1	30 300	45 400	22 100	6 660	5 500	400
35	NAST35ZZ	NAST35ZZUU	NAST35ZZR	NAST35ZZUUR	35 0 -0.012	72	25	24.8	53.5	4	1.2	1	32 200	50 600	25 700	8 130	5 000	550
40	NAST40ZZ	NAST40ZZUU	NAST40ZZR	NAST40ZZUUR	40 0 -0.012	80	26	25.8	61.5	4	1.2	1.5	35 700	61 100	30 300	9 310	4 000	710
45	NAST45ZZ	NAST45ZZUU	NAST45ZZR	NAST45ZZUUR	45 0 -0.012	85	26	25.8	66.5	4	1.2	1.5	37 100	66 400	31 100	10 100	4 000	760
50	NAST50ZZ	NAST50ZZUU	NAST50ZZR	NAST50ZZUUR	50 0 -0.012	90	26	25.8	76	4	1.2	1.5	38 700	71 800	34 000	11 000	3 500	830

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (Outside diameter) (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
NAST6,NAST8,NAST10	0/-9	0/-50
NAST12,NAST15,NAST17,NAST20	0/-11	0/-50
NAST25,NAST30,NAST35,NAST40	0/-13	0/-50
NAST45,NAST50	0/-15	0/-50

ROLLER FOLLOWERS
NON SEPARABLE WITH INNER RING

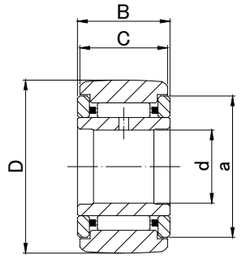




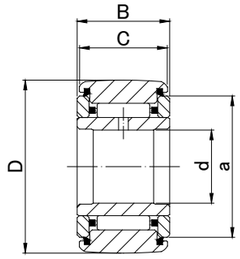
NART..R



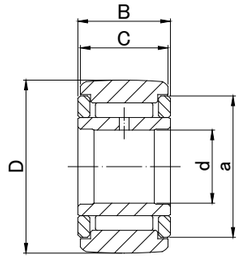
NART..VR



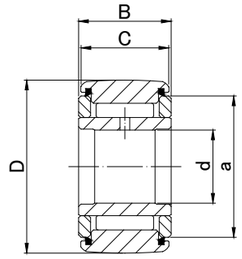
NART..R
NART..X



NART..UUR
NART..UUX



NART..VR
NART..VX



NART..UUVR
NART..UUVX

NART TYPE

Prepacked Grease

Shaft Diameter (mm)	Designation: Crowned outer ring R500(≦ NART17) R1000(NART20 ≦) Without seals	Crowned outer ring With seals	Cylindrical outer ring Without seals	Cylindrical outer ring With seals	Dimensions (mm) d	D	B	C	a	r's min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Track load capacity: Crowned outer ring N	Track load capacity: Cylindrical outer ring N	Limiting speed * rpm	Mass g (approx)
5	NART 5R NART 5VR	NART 5UUR NART 5UUVR	NART 5X NART 5VX	NART 5UUX NART 5UUVX	5 0 -0.008 5 0 -0.008	16	12	11	12	0.3	3 620 6 760	3 720 8 340	1 080	3 430	25 000 8 500	14.5 15.1
6	NART 6R NART 6VR	NART 6UUR NART 6UUVR	NART 6X NART 6VX	NART 6UUX NART 6UUVX	6 0 -0.008 6 0 -0.008	19	12	11	14	0.3	4 200 7 640	4 700 10 300	1 370	4 020	20 000 7 000	20.5 21.5
8	NART 8R NART 8VR	NART 8UUR NART 8UUVR	NART 8X NART 8VX	NART 8UUX NART 8UUVX	8 0 -0.008 8 0 -0.008	24	15	14	17.5	0.3	6 600 11 800	7 300 15 600	1 860	5 950	17 000 5 500	41.5 42.5
10	NART10R NART10VR	NART10UUR NART10UUVR	NART10X NART10VX	NART10UUX NART10UUVX	10 0 -0.008 10 0 -0.008	30	15	14	23.5	0.6	8 600 15 600	8 300 18 100	2 450	7 060	15 000 5 000	64.5 66.5
12	NART12R NART12VR	NART12UUR NART12UUVR	NART12X NART12VX	NART12UUX NART12UUVX	12 0 -0.008 12 0 -0.008	32	15	14	25.5	0.6	9 100 16 800	9 200 20 500	2 740	7 450	13 000 4 500	71 73
15	NART15R NART15VR	NART15UUR NART15UUVR	NART15X NART15VX	NART15UUX NART15UUVX	15 0 -0.008 15 0 -0.008	35	19	18	29	0.6	14 400 25 100	17 600 36 400	3 140	11 200	10 000 3 500	102 106
17	NART17R NART17VR	NART17UUR NART17UUVR	NART17X NART17VX	NART17UUX NART17UUVX	17 0 -0.008 17 0 -0.008	40	21	20	32.5	1	18 600 32 000	22 500 46 200	3 720	14 400	9 500 3 000	149 155
20	NART20R NART20VR	NART20UUR NART20UUVR	NART20X NART20VX	NART20UUX NART20UUVX	20 0 -0.010 20 0 -0.010	47	25	24	38	1	24 100 41 700	32 700 67 300	7 150	21 000	8 000 2 500	250 255
25	NART25R NART25VR	NART25UUR NART25UUVR	NART25X NART25VX	NART25UUX NART25UUVX	25 0 -0.010 25 0 -0.010	52	25	24	43	1	25 800 45 500	37 500 79 000	8 230	23 200	7 000 2 500	285 295
30	NART30R NART30VR	NART30UUR NART30UUVR	NART30X NART30VX	NART30UUX NART30UUVX	30 0 -0.010 30 0 -0.010	62	29	28	50.5	1	36 200 59 800	56 900 110 400	10 500	33 000	5 500 1 800	470 485
35	NART35R NART35VR	NART35UUR NART35UUVR	NART35X NART35VX	NART35UUX NART35UUVX	35 0 -0.012 35 0 -0.012	72	29	28	53.5	1	38 200 63 000	62 800 121 500	12 900	38 000	5 000 1 700	640 655
40	NART40R NART40VR	NART40UUR NART40UUVR	NART40X NART40VX	NART40UUX NART40UUVX	40 0 -0.012 40 0 -0.012	80	32	30	61.5	1	46 200 76 200	84 700 164 200	14 900	44 400	4 000 1 400	845 865
45	NART45R NART45VR	NART45UUR NART45UUVR	NART45X NART45VX	NART45UUX NART45UUVX	45 0 -0.012 45 0 -0.012	85	32	30	66.5	1	49 300 80 200	95 200 181 100	16 100	47 000	4 000 1 300	915 935
50	NART50R NART50VR	NART50UUR NART50UUVR	NART50X NART50VX	NART50UUX NART50UUVX	50 0 -0.012 50 0 -0.012	90	32	30	76	1	51 100 84 100	102 300 198 000	17 300	50 000	3 500 1 200	980 1 010

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (Outside diameter) (μm)

TYPE	Crowned outer ring	Cylindrical outer ring
NART5	0/-50	0/-8
NART6,NART8,NART10	0/-50	0/-9
NART12,NART15,NART17,NART20	0/-50	0/-11
NART25,NART30,NART35,NART40	0/-50	0/-13
NART45,NART50	0/-50	0/-15

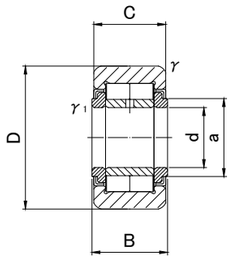
ROLLER FOLLOWERS

FULL COMPLEMENT DOUBLE ROW NON SEPARABLE WITH INNER RING





NURT



NURT..R
NURT..X

NURT TYPE

Prepacked Grease

Shaft Diameter (mm)	Designation		Dimensions (mm)							Basic dynamic load rating	Basic static load rating	Track load capacity		Limiting speed	Mass
	Crowned outer ring R500(≦NURT17-1) R1000(NURT20≦)	Cylindrical outer ring	d	D	B	C	a	*r*s min	*r*1s min	Cr N	Cor N	Crowned outer ring N	Cylindrical outer ring N	rpm	g (approx)
15	NURT15R	NURT15X	15 0 -0.008	35	19	18	20	0.6	0.3	23 400	27 200	3 140	11 200	5 200	100
	NURT15-1R	NURT15-1X	15 0 -0.008	42	19	18	20	0.6	0.3	23 400	27 200	3 930	13 900	5 200	160
17	NURT17R	NURT17X	17 0 -0.008	40	21	20	22	1	0.5	25 200	30 900	3 720	14 400	4 700	147
	NURT17-1R	NURT17-1X	17 0 -0.008	47	21	20	22	1	0.5	25 200	30 900	4 550	17 300	4 700	222
20	NURT20R	NURT20X	20 0 -0.010	47	25	24	27	1	0.5	38 900	48 900	7 150	21 000	3 800	245
	NURT20-1R	NURT20-1X	20 0 -0.010	52	25	24	27	1	0.5	38 900	48 900	8 230	23 200	3 800	321
25	NURT25R	NURT25X	25 0 -0.010	52	25	24	31	1	0.5	43 000	58 100	8 230	23 200	3 300	281
	NURT25-1R	NURT25-1X	25 0 -0.010	62	25	24	31	1	0.5	43 000	58 100	10 500	27 400	3 300	450
30	NURT30R	NURT30X	30 0 -0.010	62	29	28	38	1	0.5	57 500	74 300	10 500	32 000	2 800	466
	NURT30-1R	NURT30-1X	30 0 -0.010	72	29	28	38	1	0.5	57 500	74 300	12 900	37 200	2 800	697
35	NURT35R	NURT35X	35 0 -0.012	72	29	28	44	1.1	0.6	63 300	87 500	12 900	37 200	2 300	630
	NURT35-1R	NURT35-1X	35 0 -0.012	80	29	28	44	1.1	0.6	63 300	87 500	14 900	41 300	2 300	840
40	NURT40R	NURT40X	40 0 -0.012	80	32	30	51	1.1	0.6	86 900	124 600	14 900	44 300	1 900	817
	NURT40-1R	NURT40-1X	40 0 -0.012	90	32	30	51	1.1	0.6	86 900	124 600	17 300	49 800	1 900	1 130
45	NURT45R	NURT45X	45 0 -0.012	85	32	30	55	1.1	0.6	91 700	137 100	16 100	47 100	1 700	883
	NURT45-1R	NURT45-1X	45 0 -0.012	100	32	30	55	1.1	0.6	91 700	137 100	19 840	55 400	1 700	1 400
50	NURT50R	NURT50X	50 0 -0.012	90	32	30	60	1.1	0.6	96 300	149 700	17 300	49 800	1 500	950
	NURT50-1R	NURT50-1X	50 0 -0.012	110	32	30	60	1.1	0.6	96 300	149 700	22 530	60 900	1 500	1 690

OUTER RINGS TOLERANCE (Outside diameter) (μm)

TYPE	Crowned outer ring	Cylindrical outer ring
NURT15, NURT15-1,NURT17,NURT17-1,NURT20	0/-50	0/-11
NURT20-1,NURT25,NURT25-1,NURT30,NURT30-1,NURT35,NURT35-1,NURT40	0/-50	0/-13
NURT40-1,NURT45,NURT45-1,NURT50,NURT50-1	0/-50	0/-15

NEEDLE ROLLER BEARINGS

STAINLESS STEEL



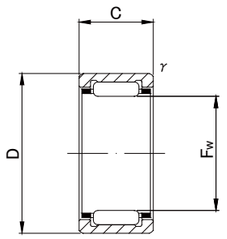
MACHINED RING NEEDLE ROLLER BEARINGS STAINLESS STEEL

WITHOUT INNER RING

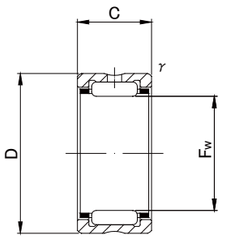
SUS/INOX



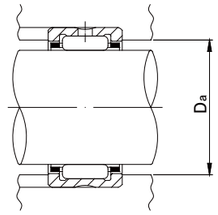
RNA49..M , NK..M



NK..M(Fw ≦ 10)



RNA49..M,NK..M



RNA49..M NK..M TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA49..M	NK..M	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
5	—	NK5/10M	5 +0.018 +0.010	10	10	0.15	6.5	2 200	1 700	40 000	3.4	—	—
	—	NK5/12M	5	10	12	0.15	6.5	2 800	2 400	40 000	4.2	—	—
6	—	NK6/10M	6 +0.018 +0.010	12	10	0.015	7.5	2 400	2 100	37 000	5.3	—	—
	—	NK6/12M	6	12	12	0.15	7.5	3 100	2 900	37 000	6.4	—	—
7	RNA495M	—	7 +0.022 +0.013	13	10	0.15	8.5	2 700	2 400	34 000	5.9	—	—
	—	NK7/10M	7	14	10	0.3	8.5	3 300	2 700	34 000	6.9	—	—
	—	NK7/12M	7 —	14	12	0.3	8.5	4 200	3 700	34 000	8.3	—	—
8	RNA496M	—	8 +0.022 +0.013	15	10	0.15	13.8	3 500	3 100	32 000	7.3	IR6810M	NA496M
	—	NK8/12M	8	15	12	0.3	13	4 600	4 300	32 000	9	IR5812M	NKI 5/12M
	—	NK8/16M	8	15	16	0.3	13	6 500	6 700	32 000	13	IR5816M	NKI 5/16M
9	—	NK9/12M	9 +0.022 +0.013	16	12	0.3	14	5 000	4 800	30 000	10	IR6912M	NKI 6/12M
	—	NK9/16M	9	16	16	0.3	14	6 900	7 500	30 000	13.2	IR6916M	NKI 6/16M
	RNA497M	—	9	17	10	0.15	15.8	4 100	3 300	30 000	9.3	IR7910M	NA497M
10	—	NK10/12M	10 +0.022 +0.013	17	12	0.3	15	5 400	5 500	28 000	10.7	IR71012M	NKI 7/12M
	—	NK10/16M	10	17	16	0.3	15	7 500	8 400	28 000	14.3	IR71016M	NKI 7/16M
	RNA498M	—	10	19	11	0.2	17.4	5 700	4 600	28 000	12.6	IR81011M	NA498M
12	—	NK12/12M	12 +0.027 +0.016	19	12	0.3	17	6 000	6 700	26 000	12.2	IR91212M	NKI 9/12M
	—	NK12/16M	12	19	16	0.3	17	8 400	10 300	26 000	16.3	IR91216M	NKI 9/16M
	RNA499M	—	12	20	11	0.3	18	6 000	5 700	26 000	13.6	IR91211M	NA499M
14	RNA4900M	—	14 +0.027 +0.016	22	13	0.3	20	8 400	9 200	24 000	16.5	IR101413M	NA4900M
	—	NK14/16M	14	22	16	0.3	20	10 800	12 600	24 000	21	IR101416M	NKI 10/16
	—	NK14/20M	14	22	20	0.3	20	13 600	17 000	24 000	26.5	IR101420M	NKI 10/20
15	—	NK15/16M	15 +0.027 +0.016	23	16	0.3	21	11 400	13 700	23 000	22.5	—	—
	—	NK15/20M	15	23	20	0.3	21	14 300	18 500	23 000	28	—	—

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

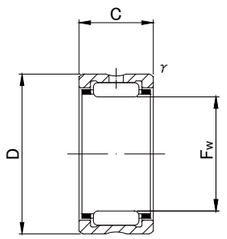
MACHINED RING NEEDLE ROLLER BEARINGS STAINLESS STEEL

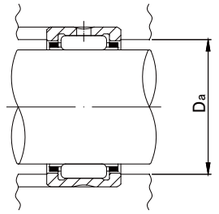
WITHOUT INNER RING

SUS/INOX



RNA49..M , NK..M





RNA49..M,NK..M

RNA49..M NK..M TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA49..M	NK..M	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
16	RNA4901M	—	16	24	13	0.3	22	8 900	10 200	23 000	18.1	IR121613M	NA4901M
	—	NK16/16M	16 +0.027 +0.016	24	16	0.3	22	11 300	13 800	23 000	23	IR121616M	NKI 12/16M
	—	NK16/20M	16	24	20	0.3	22	14 300	18 700	23 000	29	IR121620M	NKI 12/20M
17	—	NK17/16M	17 +0.027 +0.016	25	16	0.3	23	11 700	14 900	22 000	24.5	—	—
	—	NK17/20M	17	25	20	0.3	23	14 900	20 300	22 000	30.5	—	—
18	—	NK18/16M	18 +0.027 +0.016	26	16	0.3	24	12 300	16 100	21 000	25.5	—	—
	—	NK18/20M	18	26	20	0.3	24	15 600	21 700	21 000	32	—	—
19	—	NK19/16M	19 +0.033 +0.020	27	16	0.3	25	12 800	17 200	21 000	27	IR151916M	NKI 15/16M
	—	NK19/20M	19	27	20	0.3	25	16 200	23 200	21 000	34	IR151920M	NKI 15/20M
20	RNA4902M	—	20 +0.033 +0.020	28	13	0.3	26	10 000	12 600	20 000	21.5	IR152013M	NA4902M
22	RNA4903M	—	22 +0.033 +0.020	30	13	0.3	28	10 800	14 300	18 000	23.5	IR172213M	NA4903M
25	RNA4904M	—	25 +0.033 +0.020	37	17	0.3	35	19 300	23 000	16 000	55.5	IR202517M	NA4904M
30	RNA4905M	—	30 +0.033 +0.020	42	17	0.3	40	21 800	28 200	13 000	64	IR253017M	NA4905M

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.